The Definition of the Derivative

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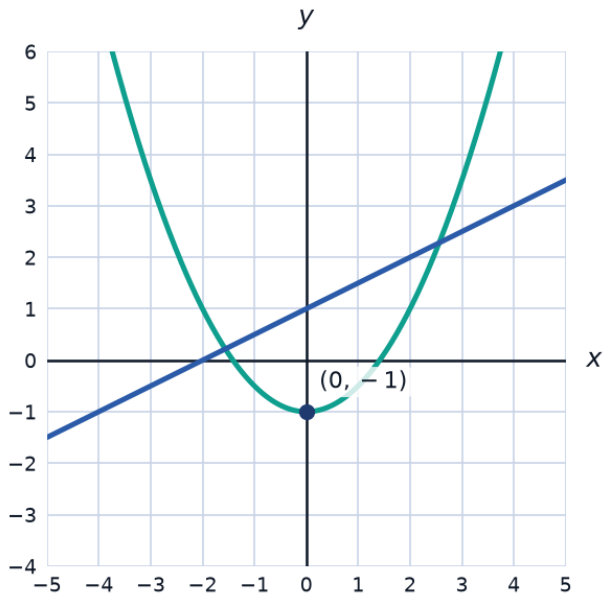
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Difference Quotients & Limit Definitions

Let $f(x)$ be a function such that $f'(x) = \cos(x^2)$ and $g(x)$ be a function defined by $g(x) = \sqrt{x}$. ; ; \textbf{(a)} Find the derivative of the composite function $h(x) = f(g(x))$. ; ; \textbf{(b)} If $f(1) = \pi$, find the value of $h'(4)$.

2 [Foundational] • Target: Differentiability Implies Continuity

Use the Second Derivative Test to classify the critical points of $f(x) = 3x^3 - 4x$.



3 [Foundational] • Target: Tangent and Normal Lines

Let $f(x) = \sin(x^2)$. What is the derivative of $f(x)$?

4 [Foundational] • Target: Estimating Derivatives from Tables & Graphs

The graph of the function $f(x) = ax - 3$ is shown below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 0$ to $x = 4$ is 20 square units. Find the value of the parameter a.

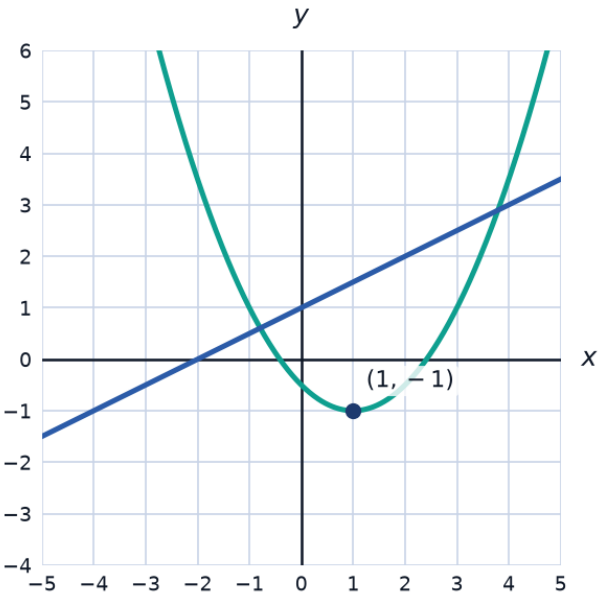
5 [Foundational] • Target: Difference Quotients & Limit Definitions

Consider the function $f(x) = x^2 - 3x$. If $g(x)$ is a function such that $h(x) = f(g(x))$, and at $x = 2$, $g(2) = 5$ and $g'(2) = 3$, find $h'(2)$.

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6 [Foundational] • Target: Differentiability Implies Continuity

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?



7 [Foundational] • Target: Tangent and Normal Lines

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

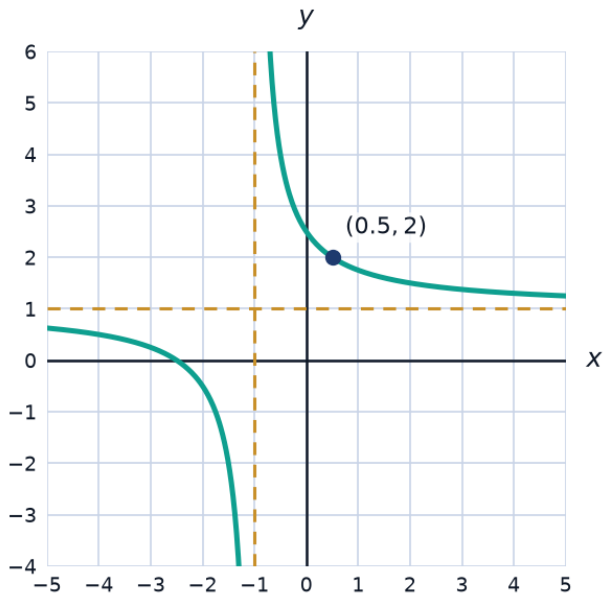
9 [Intermediate] • Target: Difference Quotients & Limit Definitions

What is the value of $\lim_{h \to 0} \frac{(2+h)^4 - 16}{h}$?

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10 [Intermediate] • Target: Differentiability Implies Continuity

Use L'Hopital's Rule to evaluate $\lim_{x \to \infty} \frac{7x}{e^{4x}}$.



11 [Intermediate] • Target: Tangent and Normal Lines

A 5-ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/s. Find the rate at which the top of the ladder is sliding down the wall when the bottom is 3 ft from the wall.

12 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

Find $\frac{d}{dx}[\sin((-2x-2)^3)]$.

13 [Intermediate] • Target: Difference Quotients & Limit Definitions

A particle's velocity is given by the function $v(t) = 5$ for $0 \le t \le 4$. Find the total displacement of the particle over this time interval.

14 [Intermediate] • Target: Differentiability Implies Continuity

Find the area of the region bounded by $y = 28x$ and $y = x^2$ in the first quadrant.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

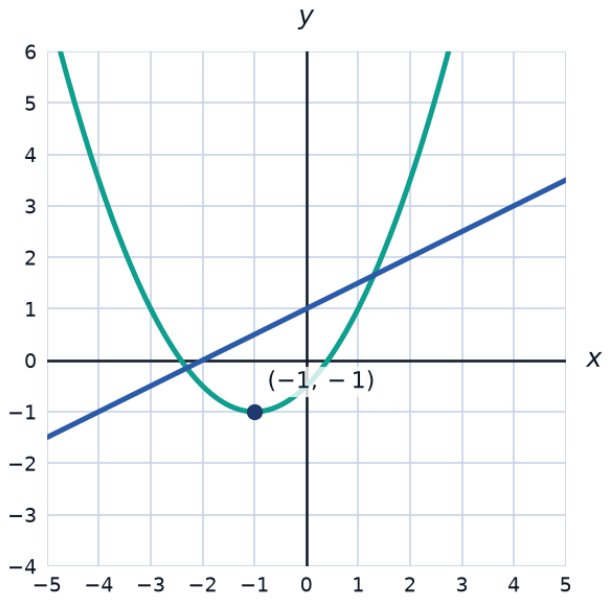
15 [Challenging] • Target: Tangent and Normal Lines • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find $\frac{d}{dx}[\sin((1-2x)^3)]$.

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16 [Challenging] • Target: Estimating Derivatives from Tables & Graphs • Cross-Lesson: 3.01 The Chain Rule

Given $\int_{-2}^{-1} 1x^2\,dx = \frac{7}{3}$ and $\int_{-1}^{0} 1x^2\,dx = \frac{1}{3}$, find $\int_{-2}^{0} 1x^2\,dx$.



17 [Challenging] • Target: Difference Quotients & Limit Definitions • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Given $\int_{-1}^{0} 4x^2\,dx = \frac{4}{3}$ and $\int_{0}^{1} 4x^2\,dx = \frac{4}{3}$, find $\int_{-1}^{1} 4x^2\,dx$.

18 [Challenging] • Target: Differentiability Implies Continuity • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

A differentiable function $f(x)$ is given by the equation $f(x) = x^2 - 6x + 10$. Find the instantaneous rate of change of $f(x)$ at $x = 4$.

19 [Challenging] • Target: Tangent and Normal Lines • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Given $\int_{-2}^{1} 5x^2\,dx = 15$ and $\int_{1}^{5} 5x^2\,dx = \frac{620}{3}$, find $\int_{-2}^{5} 5x^2\,dx$.

20 [Challenging] • Target: Estimating Derivatives from Tables & Graphs • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Given $\int_{0}^{3} 4x^2\,dx = 36$ and $\int_{3}^{5} 4x^2\,dx = \frac{392}{3}$, find $\int_{0}^{5} 4x^2\,dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Difference Quotients & Limit Definitions

Which limit expression represents the derivative $f'(x)$ by definition?

a. $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

b. $\lim_{h \to 0} \frac{f(x+h) + f(x)}{h}$

c. $\lim_{h \to \infty} \frac{f(x+h) - f(x)}{h}$

d. $\frac{f(x+h) - f(x)}{h}$

22 [Foundational] • Target: Difference Quotients & Limit Definitions

Evaluate $\lim_{h \to 0} \frac{(3+h)^2 - 9}{h}$.

a. 3

b. 6

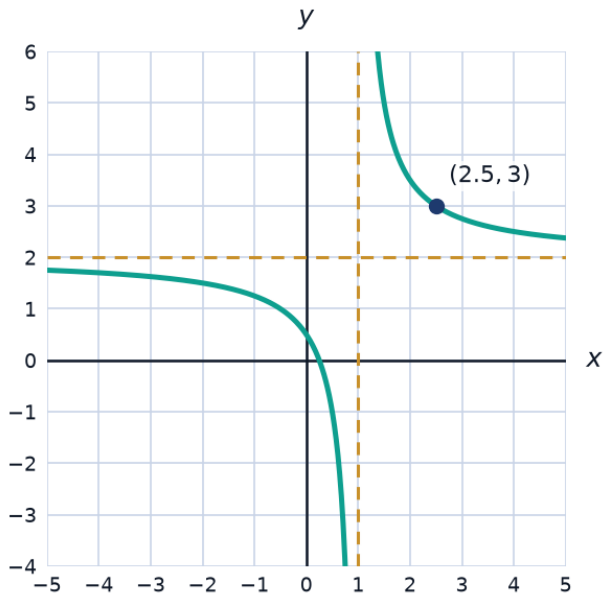
c. 9

d. 0

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23 [Foundational] • Target: Difference Quotients & Limit Definitions

Which function and value of c are represented by $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$?



a. $f(x) = \sqrt{x}$ at $c = 4$

b. $f(x) = x^2$ at $c = 2$

c. $f(x) = \sqrt{x}$ at $c = 2$

d. $f(x) = \frac{1}{\sqrt{x}}$ at $c = 4$

24 [Foundational] • Target: Differentiability Implies Continuity

If a function f is differentiable at $x = c$, which of the following statements MUST be true?

a. f is continuous at $x = c$

b. $f'(c) > 0$

c. $f''(c)$ exists

d. f has a local extremum at $x = c$

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25 [Foundational] • Target: Differentiability Implies Continuity

Why is the function $f(x) = |x - 3|$ not differentiable at $x = 3$?

a. The left-hand and right-hand derivative limits are unequal (corner)

b. $f(3)$ is undefined

c. f is not continuous at $x = 3$

d. $\lim_{x \to 3} f(x) = \infty$

26 [Foundational] • Target: Differentiability Implies Continuity

The graph of $f(x) = x^{1/3}$ has what feature at $x = 0$ that prevents differentiability?

a. A vertical tangent line where $f'(x) \to \infty$

b. A jump discontinuity

c. A sharp cusp

d. A hole

27 [Foundational] • Target: Tangent and Normal Lines

What is the equation of the tangent line to $f(x) = x^3 - 2x$ at the point $(2, 4)$?

a. $y - 4 = 10(x - 2)$

b. $y - 4 = 8(x - 2)$

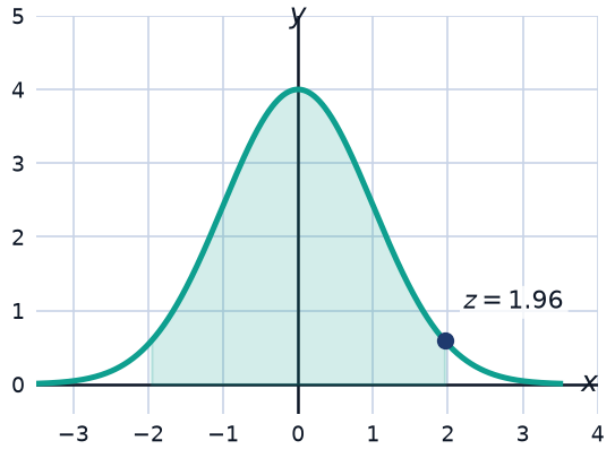
c. $y - 4 = 12(x - 2)$

d. $y = 10x - 12$

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28 [Foundational] • Target: Tangent and Normal Lines

What is the slope of the normal line to $y = x^2 + 3x$ at $x = 1$?



a. $-\frac{1}{5}$

b. 5

c. -5

d. $\frac{1}{5}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Tangent and Normal Lines

At what point on the parabola $y = x^2 - 4x + 3$ is the tangent line horizontal?

a. $(2, -1)$

b. $(2, 0)$

c. $(4, 3)$

d. $(0, 3)$

30 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

Given values $t = 2, 4, 6$ with $s(2) = 14, s(4) = 30, s(6) = 54$, estimate $s'(4)$ using the average rate of change over $[2, 6]$.

a. 10

b. 8

c. 16

d. 12

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31 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

If the graph of f has a local maximum at $x = 5$ and is differentiable, what is $f'(5)$?

a. 0

b. 1

c. Positive

d. Undefined

32 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

If $f'(x) > 0$ for all $x \in (a, b)$, what does this indicate about the graph of f on (a, b)?

a. f is strictly increasing

b. f is strictly decreasing

c. f is concave up

d. f is linear

33 [Intermediate] • Target: Difference Quotients & Limit Definitions

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

a. 5

b. 12

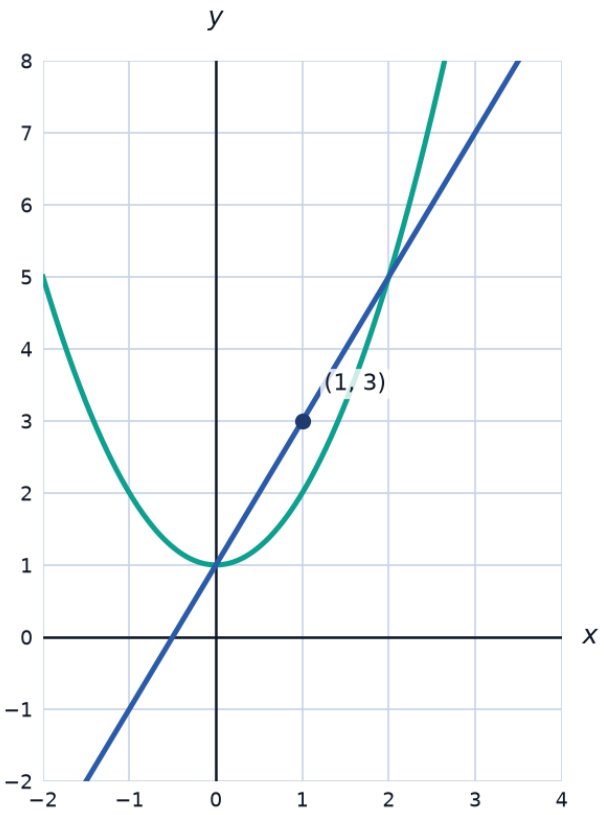
c. -3

d. 9

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34 **[Intermediate]** • **Target: Differentiability Implies Continuity**

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.



a. 1

b. 2

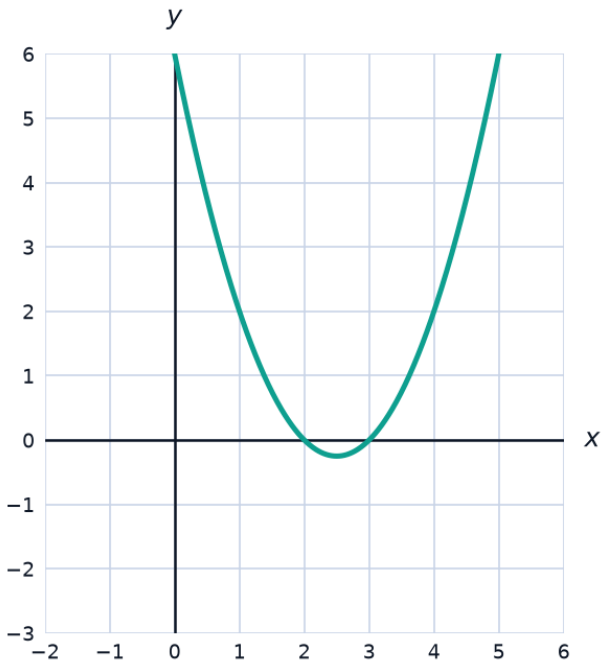
c. 3

d. 4

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35 **[Intermediate]** • **Target: Tangent and Normal Lines**

Consider the function $f(x) = x^2 - 5x + 6$. What is the limit of $f(x)$ as x approaches 2?



a. -1

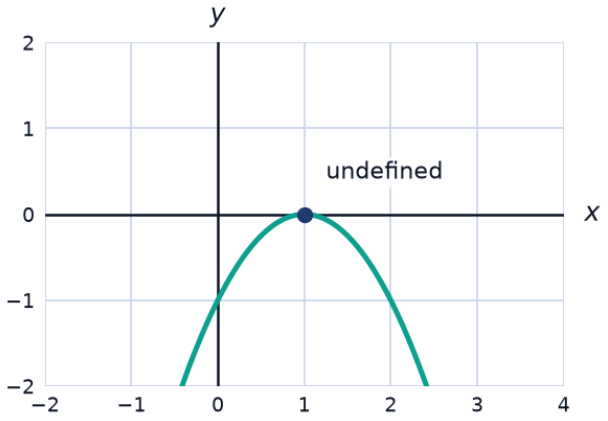
b. 0

c. 1

d. 2

36 **[Intermediate]** • **Target: Estimating Derivatives from Tables & Graphs**

Consider the function $f(x)$ graphed below. What is the limit of $f(x)$ as x approaches 1, i.e., $\lim_{x \to 1} f(x)$?



a. -1

b. 0

c. 1

d. 2

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37 [Intermediate] • Target: Difference Quotients & Limit Definitions

A region R is bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. A solid is formed by revolving this region R about the y-axis. Find the volume of this solid. \\ \\ \\ (Hint: Consider the shell method and use a substitution that relates to the derivative of the denominator of the integrand.)

a. $\pi \ln(10)$

b. $2\pi \ln(5)$

c. $\pi \ln(5)$

d. $2\pi \ln(10)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Differentiability Implies Continuity • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Consider the function $f(x)$ defined by $f(x) = \frac{x^2-9}{x-3}$ for $xeq3$. Let $g(x)$ be a function such that $g(x) = \{f(x)$, if $x < c$; k, if $x \geq c$. If $g(x)$ is continuous at $x = c$ and the limit of $g(x)$ as x approaches c from the left is 5, determine the value of k.

a. 3

b. 5

c. 9

d. 12

39 [Challenging] • Target: Tangent and Normal Lines • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The function $f(x)$ is defined piecewise as: $f(x) = \{ax^2 + 1, , x < 2; 5x - b, , x \geq 2$ If $f(x)$ is continuous at $x = 2$, and the limit of $f(x)$ as $x \to 2$ is 9, find the values of a and b.

a. a = 2, b = 1

b. a = 1, b = 1

c. a = 2, b = -1

d. a = 1, b = -1

40 [Challenging] • Target: Estimating Derivatives from Tables & Graphs • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

The slope of the tangent to $y = x^3$ at $x = 2$ is:

a. 6

b. 8

c. 12

d. 16

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41 [Challenging] • Target: Difference Quotients & Limit Definitions • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find the derivative of $f(x) = 4e^x$.

a. e^x

b. $4e^x$

c. $-4e^x$

d. $4e^x + 4$

42 [Challenging] • Target: Differentiability Implies Continuity • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find the derivative of $f(x) = 5e^x$.

a. $-5e^x$

b. $5e^x$

c. $5e^x + 5$

d. e^x

43 [Challenging] • Target: Tangent and Normal Lines • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find the derivative of $f(x) = 3e^x$.

a. $-3e^x$

b. $3e^x + 3$

c. e^x

d. $3e^x$

44 [Challenging] • Target: Estimating Derivatives from Tables & Graphs • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the derivative of $f(x) = 1\sec x$.

a. $\tan(x)\sec(x)$

b. $\sec(x)$

c. $\tan(x)\sec(x) + 1$

d. $-\tan(x)\sec(x)$

45 [Challenging] • Target: Difference Quotients & Limit Definitions • Cross-Lesson: 8.01 Area Between Curves

Find the derivative of $f(x) = 5\cos x$.

a. $5\cos(x)$

b. $5\sin(x)$

c. $-\sin(x)$

d. $-5\sin(x)$

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